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Seeking a Spotless Mind: Extinction, Deconsolidation, and Erasure of Fear Memory

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Abstract

Learning to contend with threats in the environment is essential to survival, but dysregulation of memories for traumatic events can lead to disabling psychopathology. Recent years have witnessed an impressive growth in our understanding of the neural systems and synaptic mechanisms underlying emotional memory formation. As a consequence, interest has emerged in developing strategies for suppressing, if not eliminating, fear memories. Here I review recent work employing sophisticated behavioral, pharmacological, and molecular tools to target fear memories, placing these memories firmly behind the crosshairs of neurobiologically informed interventions.

Learning to fear threats in the environment is highly adaptive; it allows animals, whether rats or humans, to anticipate harm and organize appropriate defensive behaviors in response to threat (Bolles, 1970; Fanselow and Lester, 1988; Ohman and Mineka, 2001). However, this form of learning can also lead to pathological fear memories that fuel disorders of fear and anxiety, such as panic disorder and post-traumatic stress disorder (PTSD) in humans (Bouton et al., 2001; Rasmussen and Charney, 1997; Wolpe and Rowan, 1988). What determines an individual's vulnerability to developing pathological fear after a traumatic experience is not clear; simply experiencing trauma does not appear to be sufficient (Jovanovic and Ressler, 2010; Yehuda and Ledoux, 2007). Less than 10% of individuals that experience a traumatic event, such as a natural disaster, will develop symptoms of PTSD. Nonetheless, in those individuals that develop PTSD, an important clinical concern is how to limit pathological fear once it has been established (Powers et al., 2010; Rothbaum and Davis, 2003).

Yet limiting pathological fear is a considerable challenge insofar as fear memories are evolutionarily programmed to be rapidly acquired, temporally enduring, and broadly generalized across both familiar and novel contexts. Moreover, behavioral methods to reduce fear and anxiety, such as exposure therapy, tend to produce fear suppression that is often slow to develop, short-lived, and context-dependent (Bouton, 1988; Craske et al., 2008; Hermans et al., 2006). Apparently, the extinction learning that occurs when overt cues or mental images associated with trauma are presented alone in a safe and therapeutic setting only temporarily suppresses the dominant memory that trauma-related stimuli are fearful. This stands to reason when one considers that an incorrect attribution of safety to a

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dangerous place, object, or animal may result in death or injury, whereas an incorrect attribution of danger to an otherwise safe stimulus protects one from harm.

Given the resilience of fear memory, there has been considerable interest in understanding the neurobiological mechanisms that mediate the long-term storage and retrieval of fear memories, as well as the mechanisms underlying the safety memories acquired during extinction. Fortunately, there are several rich experimental animal models that have been developed to study emotional learning and memory, and these have yielded considerable new information concerning the neurobiology of fear conditioning and extinction. The purpose of this review is to consider how these models have contributed to recent advances in understanding the molecular and cellular mechanisms and neural circuits in the brain involved in learning new fears and inhibiting, even erasing, old ones.

Neural Substrates of Fear Conditioning

The quintessential model for the neuroscientific study of aversive learning and memory is Pavlovian fear conditioning. In this form of learning, an innocuous stimulus (conditioned stimulus or CS), such as a tone or light, is paired with a noxious stimulus (unconditioned stimulus or US), such as an electric footshock. After one or more such trials, animals rapidly learn that the CS predicts the aversive US and consequently produce a learned fear response (conditioned response or CR) to the CS. This form of learning is ubiquitous in the animal kingdom, and is now routinely used in mice, rats, cats, rabbits, primates, and humans to probe the neural systems and cellular mechanisms underlying emotional learning and memory. Importantly, nearly a century of fundamental work by experimental psychologists on the behavioral processes involved in associative learning have established conditioning methods as sophisticated tools for disentangling the brain mechanisms of sensation, memory, and action (Fanselow and Poulos, 2005).

Decades of research into the neural substrates of Pavlovian fear conditioning has revealed the essential neural circuit required for the acquisition and expression of fear memory (Figure 1) (Davis, 2006; LeDoux, 2000; Maren, 2001; Pape and Pare, 2010). The core of this fear circuit is centered on the amygdala, which is a heterogeneous collection of nuclei buried deep within the temporal lobe. Two regions within the amygdala are particularly important for fear conditioning: the basolateral complex (BLA, consisting of lateral, basolateral, and basomedial nuclei) and the central nuclear group [CEA, consisting of the medial (CEm) and lateral (CEl) nuclei]. Anatomically, the BLA represents a nuclear extension of the temporal neocortex and the CEA represents a ventrocaudal extension of the striatum (Pitkanen et al., 1997; Swanson and Petrovich, 1998). The flow of information between the BLA and CEA is largely unidirectional with LA neurons projecting to CEl directly, and indirectly to CEm via the basolateral nucleus (BL) and through a network of inhibitory interneurons in the intercalated cell masses (ITC) (Krettek and Price, 1978; Pare and Smith, 1993, 1998; Pare et al., 1995). CEm projects to several brain regions that mediate fear responses, such as freezing, tachycardia, and stress hormone release, and axonal projections from BLA to CE are critical for the expression of these responses after fear conditioning (Ciocchi et al., 2010; Haubensak et al., 2010; Jimenez and Maren, 2009; Paré et al., 2004). That is, after fear conditioning, it has recently been shown that aversive CSs come to suppress the inhibitory influence of CEl on CEm and drive the expression of conditional fear responses (Ciocchi et al., 2010; Haubensak et al., 2010). This reveals that CEl normally inhibits CEm and the regulation of this inhibition appears to be essential for the expression of fear and anxiety (Tye et al., 2011).

Multimodal sensory information reaches both regions of the amygdala, and this affords an opportunity for the convergence of CS and US information within these areas. Indeed,

substantial data indicate that the lateral nucleus (LA) is a critical sensory interface of the amygdala that mediates CS-US association formation during fear conditioning (Blair et al., 2001; Maren, 1999). For example, auditory and somatic stimuli excite LA neurons at short latencies (Johansen et al., 2010; Romanski et al., 1993), and fear conditioning greatly augments responses of LA neurons to auditory CSs (Goosens et al., 2003; Goosens and Maren, 2004; Herry et al., 2008b; Hobin et al., 2003; Johansen et al., 2010; Maren, 2000; Quirk et al., 1997; Repa et al., 2001). Bernstein and colleagues have also recently shown that individual LA neurons exhibit increases in the expression of the immediate early gene *Arc* that reflects CS-US convergence in these cells (Barot et al., 2009). The convergence of CS and US information in the LA engenders associative plasticity that increases the efficacy of CS inputs onto LA neurons (Blair et al., 2001; Maren, 1999). For example, fear conditioning increases CS-evoked extracellular field potentials in the LA *in vivo* (Rogan and LeDoux, 1995; Tang et al., 2001), and LA neurons exhibit conditioning-related changes in synaptic transmission measured *ex vivo* (McKernan and Shinnick-Gallagher, 1997; Rumpel et al., 2005; Tsvetkov et al., 2002).

In addition to these electrophysiological correlates of conditioning, LA neurons exhibit changes in gene expression and protein phosphorylation after fear conditioning (Lamprecht et al., 2009; Ploski et al., 2010; Ploski et al., 2008; Shumyatsky et al., 2005). Indeed, fear conditioning regulates several proteins importantly involved in the induction and maintenance of synaptic plasticity, including AMPA receptors, PI-3 kinase, and A-kinase anchoring proteins, nexin protease, and mitogen-activated protein kinase among others (Apergis-Schoute et al., 2005; Lamprecht et al., 2002; Lin et al., 2001; Meins et al., 2010; Migues et al., 2010; Moita et al., 2002; Schafe et al., 2000). Accordingly, inhibiting synaptic plasticity in the LA with a variety of manipulations including NMDA receptor antagonists (Fendt, 2001; Goosens and Maren, 2003; Lee and Kim, 1998; Maren et al., 1996b; Miserendino et al., 1990), protein synthesis inhibitors (Maren et al., 2003; Nader et al., 2000; Schafe and LeDoux, 2000), protein kinase inhibitors (Apergis-Schoute et al., 2005; Goosens et al., 2000; Lamprecht et al., 2002; Lin et al., 2001; Merino and Maren, 2006; Migues et al., 2010), or antisense oligonucleotides for plasticity-related genes (Malkani et al., 2004; Ploski et al., 2008) impairs the acquisition of fear memory. Similarly, inactivating LA neurons prevents fear expression (Helmstetter and Bellgowan, 1994; Maren et al., 2001; Muller et al., 1997). Genetically modified mice that lack proteins essential for amygdala synaptic plasticity exhibit deficits in fear conditioning (Brambilla et al., 1997; Shumyatsky et al., 2002).

Yet despite the focus on synaptic plasticity in the LA as a mechanism for fear conditioning, emerging evidence suggests that fear conditioning is likely mediated by distributed synaptic plasticity within the amygdala (Wilensky et al., 2006; Zimmerman et al., 2007). For instance, central nucleus neurons exhibit conditioning-related plasticity in spike firing and changes in gene expression after conditioning (Ciocchi et al., 2010; Pascoe and Kapp, 1985). Moreover, NMDA receptor antagonism in the CEA prevents the acquisition of conditioned fear (Goosens and Maren, 2003) and blocks synaptic plasticity in CEA neurons (Lopez de Armentia and Sah, 2007; Samson and Paré, 2005). Inhibition of protein synthesis in the CEA prevents the consolidation of fear memory (Wilensky et al., 2006). Within the amygdala, the CEA receives its primary excitatory input from the basolateral nucleus (BL), which in turn receives projections from LA. Hence, a cascade of NMDA receptor-dependent plasticity among these glutamatergic synapses may be essential for fear conditioning (Maren, 2008). Plasticity among inhibitory neurons in the intercalated cell masses that are interposed between the LA and CEA may also functionally disinhibit LA neurons to promote fear expression (Amano et al., 2010; Royer and Pare, 2003).

It is apparent that a considerable amount of work implicates the amygdala in fear memory. Yet which population of neurons in the amygdala is essential for fear conditioning and where are they located? This question is beginning to be addressed by powerful new molecular genetic methodologies that allow visualization of neurons involved in encoding and retrieving fear memory. For example, Mayford and colleagues (2007) have used a transgenic mouse (TetTag) expressing a doxycycline-insensitive tetracycline-transactivator (tTA*) coupled to a tauLacZ reporter to visualize neurons activated by a fear conditioning experience (Reijmers et al., 2007). In this mouse, activation of the tTA* requires a standard doxycycline-sensitive tTA, which is under the control of the immediate early gene (IEG) *Fos* promoter to index neuronal activity. Hence, once activated (off doxycycline), the tTA* transgene drives tauLacZ reporter expression even in the presence of doxycycline. This allows neurons that were active in a particular time window (when animals are off doxycycline) to be persistently tagged. Combining this method with immunohistochemistry for Zif (an IEG protein that also indexes activity), the authors were able to determine whether neurons active at the time of conditioning (tauLacZ -positive) were also active at the time of memory retrieval (Zif-positive) three days after conditioning. Indeed, the authors found that a significant subset (roughly 12%) of neurons in the BLA (CEA was not reported) co-expressed tauLacZ and Zif, and that the number of co-labeled neurons correlated with the expression of fear. Similarly, Josselyn and colleagues (2007) found that CREB-overexpressing neurons in LA were preferentially incorporated into the fear memory network insofar as those neurons were more likely to co-express Arc (an IEG protein that also indexes activity) upon memory retrieval (Han et al., 2007). Hence, these approaches have the potential to define specific neuronal networks involved in encoding and retrieving fear memories.

Once acquired, the amygdala has a long-term role in maintaining fear memory. Unlike hippocampal-dependent memories that undergo systems consolidation in neocortex (Bontempi et al., 1999; Frankland and Bontempi, 2005; Frankland et al., 2004; Squire and Alvarez, 1995), CS-US associations encoded in the amygdala appear to reside there permanently. Postconditioning lesions of the BLA yield equivalent impairments in conditional fear independent of when they are made after training (Cousens and Otto, 1998; Lee et al., 1996; Maren et al., 1996a). In fact, BLA lesions impair fear memory when made up to one year after conditioning (Gale et al., 2004), suggesting that the amygdala maintains fear memory for the life of the animal. Although we do not yet understand the nature of the permanent changes in brain circuitry that maintain fear memory for the life of the organism, we now have an anatomical locus to target interventions to either suppress or reverse fear memories.

Extinction and the Contextual Regulation of Fear

During Pavlovian conditioning, animals learn that a CS predicts the occurrence of a US. This predictive association fosters adaptive, anticipatory learned responses when the CS occurs. Given that conditioning depends on the CS-US contingency, it follows that breaking the contingency might reduce conditional responding. To break the CS-US contingency, Pavlov developed an experimental procedure in which the CS was presented alone (without the US) for several trials after the completion of conditioning (Pavlov, 1927). Not surprisingly, the earliest CS-alone trials produced a robust CR, but the CR gradually faded with subsequent CS presentations. Pavlov termed this phenomenon “extinction”, and it is now apparent that this form of learning is an important component of behavioral interventions for patients with pathological fear memories. For example, exposure therapy involves the use of mental imagery and exposure to trauma-relevant cues in a safe environment to suppress the fear associated with the memory of the traumatic event (Craske et al., 2008; Powers et al., 2010; Rothbaum and Davis, 2003).

Given the importance of extinction learning as a mechanism for suppressing fear memory, there has been an explosion of work into the neural mechanisms of extinction (Bouton et al., 2006a; Herry et al., 2010; Myers and Davis, 2002; Pape and Pare, 2010; Quirk and Mueller, 2008). Not surprisingly, much of this work has focused on the contribution of the amygdala to fear extinction and several reports indicate that the BLA is critical for the acquisition of extinction memories. For example, infusing NMDA receptor antagonists into the BLA disrupts the acquisition of extinction (Falls et al., 1992; Laurent et al., 2008; Zimmerman and Maren, 2010), whereas blockade of NMDA receptors in the CEA does not affect extinction learning (Zimmerman and Maren, 2010). Intracellular signaling pathways downstream of BLA NMDA receptors are also critical for extinction learning (Herry et al., 2006; Lin et al., 2003a; Lin et al., 2003c; Lu et al., 2001; Yang and Lu, 2005). In addition to the glutamatergic system, recent work indicates that other neurotransmitter systems contribute to extinction learning. For example, mice lacking endocannabinoid receptors (CB1 receptors, specifically) exhibit impairments in extinction learning and systemic administration of a CB1 antagonist (SR141716, rimonabant) inhibits extinction learning (Chhatwal et al., 2009; Marsicano et al., 2002). Endocannabinoids modulate inhibitory GABAergic synaptic transmission in the amygdala, which is also essential for extinction learning (Chhatwal et al., 2005b; Harris and Westbrook, 1998; Laurent et al., 2008; Laurent and Westbrook, 2008; Makkar et al., 2010). Collectively, these data suggest that changes in the synaptic transmission within the BLA contribute to the suppression of conditional fear after extinction training. Indeed, depotentiation of amygdaloid synaptic transmission has been reported to occur after extinction training (Kim et al., 2007).

Curiously, however, extinction produces a relatively transient suppression of fear; conditioned fear responses return under a variety of conditions including after the mere passage of time (i.e., spontaneous recovery) or if the extinguished CS is presented outside the extinction context (e.g., renewal) (Bouton, 1993). This suggests that memories of both fear conditioning and extinction are encoded in the amygdala, and contextual retrieval cues determine which memory is expressed in behavior. The medial prefrontal cortex and hippocampus have rich connections with the amygdala and are involved in processing contextual information. Not surprisingly, considerable work now implicates these brain areas in the regulation of fear expression after extinction (Maren and Quirk, 2004; Quirk et al., 2006; Quirk and Mueller, 2008; Quirk et al., 2000; Sotres-Bayon et al., 2006; Sotres-Bayon and Quirk, 2010).

Anatomically, the infralimbic (IL) division of the vmPFC projects to a network of inhibitory interneurons in the amygdala; these neurons are located in the intercalated cell masses (ITC) interposed between the BLA and CEA (Figure 1). ITC neurons send massive inhibitory projections to CEA, and are therefore well positioned to limit excitatory input from the BLA and reduce CEA-mediated fear responses (Berretta et al., 2005; Paré and Smith, 1993). Paré and colleagues recently demonstrated the important role for ITC neurons in the expression of extinction using selective lesions of ITC neurons (Likhtik et al., 2008). In this study, rats received intra-amygdala infusions of a selective immunotoxin against ITC neurons after extinction training; ITC lesions produced a significant loss of extinction (i.e., the expression of freezing was increased by the lesion). Other work has shown that pharmacological manipulation of the vmPFC influences the consolidation of extinction memory (Hugues et al., 2006; Laurent and Westbrook, 2008, 2009; Mueller et al., 2010; Sierra-Mercado et al., 2011), suggesting that the vmPFC may also play a role in establishing extinction memories (as opposed to merely regulating extinction recall). Extinction learning and recall induces Fos in vmPFC neurons (Hefner et al., 2008; Herry and Mons, 2004; Knapska and Maren, 2009) and electrical stimulation of the vmPFC facilitates extinction (Milad et al., 2004; Vidal-Gonzalez et al., 2006). Prefrontal cortical neurons also exhibit physiological changes, including increased bursting, during extinction learning (Burgos-Robles et al., 2007; Chang

et al., 2010; Milad and Quirk, 2002). Interestingly, several studies report intact extinction after vmPFC lesions (Farinelli et al., 2006; Garcia et al., 2006; Gewirtz et al., 1997); some of these disparities may arise from strain differences in the effects of PFC lesions on extinction (Chang and Maren, 2010).

In addition to the mPFC, the hippocampus has been implicated in both the acquisition and expression of fear extinction (Bouton et al., 2006b). A number of groups have shown that disrupting hippocampal function prior to extinction training impairs the acquisition of extinction (Corcoran et al., 2005; Fischer et al., 2007; Fischer et al., 2004; Sananbenesi et al., 2007; Szapiro et al., 2003; Tronson et al., 2009; Vianna et al., 2001). Moreover, the hippocampus is involved in the context-dependence of extinction, particularly for regulating the renewal of fear to an extinguished CS outside of the extinction context (Maren and Holt, 2000). Lesions or reversible inactivation of the hippocampus prevents the renewal of fear to an extinguished CS outside of the extinction context (Corcoran et al., 2005; Corcoran and Maren, 2001, 2004; Hobin et al., 2006; Ji and Maren, 2008a; Ji and Maren, 2005), and a recent report implicates electrical synapses in the hippocampus in this function (Bissiere et al., 2011). Both cortical and subcortical projections of the hippocampus are important for renewal insofar as fornix or entorhinal cortical lesions reproduce the effects of hippocampal lesions (Ji and Maren, 2008b). Inactivation of the hippocampus also eliminates “neuronal renewal” of CS-evoked neuronal activity that is observed in the LA when an extinguished CS is presented outside the extinction context (Hobin et al., 2003; Maren and Hobin, 2007). Collectively, these data reveal that extinction yields a new inhibitory memory that competes with the fear memory for expression in behavior. Disruption of the hippocampal system prevents the return of fear normally observed when an extinguished CS is presented outside of the extinction context. Animals without a functional hippocampus are unable to contextualize their fear and extinction memories and therefore respond according to the net experience with the CS. Hence, disrupting hippocampal function actually promotes the generalization of extinction memories to many contexts.

The hippocampus projects to both the BLA and the vmPFC and is therefore well positioned to gate the expression of fear and extinction memories. Indeed, a recent study in our laboratory showed that different prefrontal-amygdala circuits were engaged during the retrieval of fear and extinction memories (Knapska and Maren, 2009). Figure 2 highlights the brain regions (in red) that exhibited differential c-fos expression during retention tests in which an extinguished CS was presented either in the extinction context or in another context. Interestingly, neurons in the IL and ITC were active during the retrieval test in the extinction context, when conditioned freezing was suppressed, but not outside the extinction context when conditioned fear was high. Conversely, neurons in the PL, LA, and CEM were active outside the extinction context (when fear was high), but not in the extinction context (when fear was low). The hippocampus was engaged under both conditions, suggesting that it may uniquely process where and when a CS is experienced, independent of the valence of that memory.

It is possible that contextual processing by the hippocampus gates fear through direct projections to the amygdala, or through indirect projections through vmPFC. To this point, recent work by Luthi and colleagues has shown that there are anatomically distinct populations of neurons in the basolateral amygdaloid nucleus that respond when animals express either express or suppress conditional fear (Herry et al., 2008a). “Fear” neurons responded to non-extinguished CSs or extinguished CSs presented outside the extinction context, whereas extinction neurons only responded to extinguished CSs presented in their extinction context. Interestingly, the majority of fear neurons were orthodromically activated by electrical stimulation of the ventral hippocampus, whereas extinction neurons received their afferent input from the vmPFC. Hence, the contextual retrieval of fear memory might

involve a hippocampo-prefrontal cortical network that regulates the balance of excitation and inhibition in the amygdala to foster or suppress, respectively, fear to an extinguished CS (Maren, 2005). It is also conceivable that the balance of activity among inhibitory CEI neurons that are either excited (“CS on” neurons) or inhibited (“CS off” neurons) by a CS (e.g., Ciochi et al., 2010) regulates the suppression or renewal, respectively, of fear after extinction; this possibility has not yet been explored.

Preventing the Return of Fear

Reducing the expression of fear memory with extinction procedures, such as exposure therapy, is fundamental to therapeutic interventions for fear and anxiety disorders in humans. Unfortunately, the suppression of conditional responding that follows extinction is transient (Bouton, 1993; Bouton and Bolles, 1979a; Rescorla, 2004). In his early work, Pavlov noted that an extinguished CR would return if the animal was presented with a novel stimulus, a phenomenon termed “disinhibition” (Pavlov, 1927). He also showed that extinguished CRs would spontaneously return with the mere passage of time, a phenomenon termed “spontaneous recovery”. As previously described, extinguished CRs are also highly specific to the experimental context in which they are acquired. In other words, an extinguished CR exhibits “renewal” when the CS is presented outside the extinction context. Similarly, unsignaled USs can restore extinguished responding when the CS is presented in the context in which the US was delivered. This phenomenon is termed “reinstatement” (Bouton and Bolles, 1979b; Rescorla and Heth, 1975). These phenomena indicate that extinction does not erase the conditioning memory, rather it causes new learning about the CS. Indeed, it appears that extinction training yields a new “safety” memory that inhibits retrieval of the fear memory. Unlike fear memory, the expression of this safety memory is limited by context and time (Bouton, 1993). “Context” is defined broadly to include the experimental environment and interoceptive state of the animal, as well as the actual (time of day) and relative time (how long ago) the events were learned. A challenge then for clinicians is to develop therapeutic interventions that yield fear suppression not only in the treatment context, but also in every context in which a fear stimulus is encountered. A challenge for neuroscientists is to understand not only how safety memories are encoded alongside fear memories in the brain, but also how these different memories come to be regulated by time and context.

Given that considerable progress has been made in elucidating the neural substrates of extinction, several investigators have examined whether extinction learning can be facilitated. Some of the early work in this arena focused on the FDA-approved drug, D-cycloserine (DCS), which is an allosteric modulator of the NMDA receptor that facilitates agonist binding therefore increasing NMDA receptor function. It has been reported that either systemic or intra-amygdaloid administration of DCS facilitates extinction learning (Ledgerwood et al., 2003, 2005; Walker et al., 2002). Interestingly, there have been reports that the extinction memory acquired under DCS is less likely to exhibit recovery (e.g., reinstatement) (Ledgerwood et al., 2004), although this outcome has not been universally reported (Bouton et al., 2008; Woods and Bouton, 2006). Moreover, administering DCS prior to extinction in rats appears to promote the reversal of some of the synaptic changes in the lateral amygdala that accompany fear conditioning (Mao et al., 2008). Preliminary work using DCS as a pharmacological adjunct to exposure therapy has shown some promise (Davis et al., 2006). For example, administration of DCS along with controlled exposure therapy improved outcomes for patients with fear of heights (Ressler et al., 2004) and social anxiety disorder (Guastella et al., 2008), although it did not improve therapeutic outcomes for spider phobics (Guastella et al., 2007a) or effect the extinction of fear conditioning (Guastella et al., 2007b).

Another compound that has been reported to enhance extinction learning is yohimbine, an alpha2-adrenergic agonist that has been used in humans to treat erectile dysfunction. Cain and colleagues reported that systemic yohimbine administration prior to extinction training in mice increased the long-term retention of extinction the following day (Cain et al., 2004). However, the effects of yohimbine on extinction are quite variable, in some cases even impairing extinction learning (Holmes and Quirk, 2010). Nonetheless, a recent report in humans suggests that yohimbine administration enhances the efficacy of exposure therapy in claustrophobic patients (Powers et al., 2009). The role for the adrenergic system in extinction learning is likely to be quite complex, however, insofar as prazosin, an alpha1-adrenoceptor antagonist, has recently been reported to impair fear extinction in mice after systemic (Bernardi and Lattal, 2010; Do-Monte et al., 2010) and intra-vmPFC administration (Do-Monte et al., 2010). Prazosin is currently used in the treatment of nightmares and sleep disorders in patients with PTSD (Raskind et al., 2007; Van Liempt et al., 2006), but its effect on extinction learning suggests that it may have limited efficacy as an adjunct to exposure therapy.

Endocannabinoids provide another potential route for enhancing extinction (Lutz, 2007). CB1 receptors are localized on inhibitory interneurons in the amygdala (Azad et al., 2004), and may regulate the activity of these neurons during extinction learning (Chhatwal et al., 2005a; Chhatwal et al., 2009). Systemic administration of drugs that enhance cannabinoid signaling, such as the reuptake inhibitor AM404 and the CB1 receptor agonist WIN55212-2, have been reported to facilitate extinction learning under some conditions (Marsicano et al., 2002), although chronic administration of WIN55212-2 has recently been reported to impair extinction learning (Lin et al., 2008). Moreover, there is recent data suggesting that CB1 receptors may not have a specific role in long-term fear extinction, but may be more generally involved in behavioral habituation (Plendl and Wotjak, 2010). These drugs have not been approved for use in humans, however, so it is not known whether increasing activity at endocannabinoid receptors would facilitate exposure therapy, for example.

Recently, Quirk and colleagues have reported that they can produce a pharmacologically-induced extinction without any behavioral training (Peters et al., 2010). They infused brain-derived neurotrophic factor (BDNF) into the infralimbic cortex twenty-four hours after fear conditioning and found that the expression of fear to the auditory CS was greatly diminished the following day. A series of control experiments ruled out the possibility that the infusion disrupted performance or the fear memory itself; notably, the fear memory was readily reinstated by additional unsignaled footshock. Analysis of BDNF levels in brain revealed animals that successfully extinguished fear showed elevated levels of BDNF in the hippocampus. Hippocampal infusions of BDNF were found to reproduce the effects of IL BDNF infusions, and infusing a BDNF-sequestering antibody into the IL disrupted this effect. These results extend other studies that have implicated BDNF in extinction learning (Chhatwal et al., 2006) and may explain why genetic variation in the gene encoding BDNF correlates with extinction in humans (Soliman et al., 2010). Indeed, they reveal a novel pharmacological target for either enhancing fear extinction during exposure therapy or even inducing fear extinction without formal exposure therapy. Ultimately, combining behavioral strategies to optimize extinction learning (Craske et al., 2008) with pharmacological adjuncts such as BDNF or DCS may yield even greater fear suppression in patients with anxiety disorders than has been achieved with traditional therapeutic interventions.

Erasing the Trace

Despite the promise of enhancing extinction with behavioral and pharmacological approaches, the lability of extinction memory suggests that, at best, it can only temporarily quell traumatic fear. A more effective way to eliminate unwanted fears would be to erase the

fear memory itself. It has long been appreciated that new memories undergo a period of consolidation in which they are labile and sensitive to disruption (McGaugh, 2000). Long-term synaptic plasticity in the brain requires *de novo* protein synthesis (Deadwyler et al., 1987; Krug et al., 1984; Stanton and Sarvey, 1984), and administration of protein synthesis inhibitors soon after learning produces memory impairments (Agranoff et al., 1965; Agranoff and Klinger, 1964; Davis and Squire, 1984). Therefore, one strategy for reducing pathological fear would be to prevent the consolidation of long-term fear memories soon after a traumatic experience. Consistent with this aim, several investigators have now shown that fear memory is inhibited by systemic post-training protein synthesis inhibition (Bourtchouladze, 1998; Lattal and Abel, 2001). Moreover, infusion of the protein synthesis inhibitor anisomycin into the BLA within hours of fear conditioning disrupts the consolidation of long-term fear memories and reduces conditional fear responses (Maren et al., 2003; Parsons et al., 2006; Schafe and LeDoux, 2000; Schafe et al., 1999).

Immediate extinction

In addition to protein synthesis inhibitors, administering behavioral interventions soon after fear conditioning might also disrupt long-term fear memory by interfering with consolidation processes. For example, it has been reported that administering low-frequency stimulation soon after fear conditioning eliminates conditioning-related changes in MAPK phosphorylation in the BLA, a biochemical correlate of long-term synaptic plasticity and fear memory, as well as fear memory (Lin et al., 2003b). Based on this evidence, Davis and colleagues explored whether administering extinction trials soon after fear conditioning would yield a permanent loss of fear, rather than the temporary inhibition of fear typically observed with delayed extinction training (Myers et al., 2006). To test this, they administered extinction trials shortly (i.e., 10 min) after fear-potentiated startle conditioning in rats and examined whether fear suppression was more durable than that produced by extinction 24 hours after conditioning. They found that this immediate extinction procedure resulted in a loss of fear that was quite durable and exhibited little spontaneous recovery, reinstatement, or renewal. The implication of these findings is that early extinction training resulted in a permanent fear loss, which is not typical when extinction training is conducted one day after conditioning. The lack of fear recovery in this report suggested that immediate extinction might disrupt the consolidation of fear memory, yielding a relatively permanent loss of fear.

Although promising, several laboratories have now found that immediate extinction does not always reduce the recovery of fear (Archbold et al., 2010; Huff et al., 2009), and in fact may not suppress fear at all under some conditions (Chang and Maren, 2009; Maren and Chang, 2006; Woods and Bouton, 2008). For example, we have found that extinction training soon after fear conditioning produces short-term suppression of conditional freezing during the extinction session, but this suppression is not long lasting and fully recovers the following day (Maren and Chang, 2006). In fact, recently acquired fear memories appear to be particularly resistant to extinction insofar as we failed to obtain long-term fear loss even after over 200 extinction trials. In our hands, this “immediate extinction deficit” was obtained up to 6 hours after fear conditioning (Chang and Maren, 2009), suggesting that there is a substantial time window after fear conditioning in which fear memory is resistant to extinction. Interestingly, two recent papers suggest that immediate extinction does not engage medial prefrontal cortical circuits involved in extinction learning (Chang et al., 2010; Kim et al., 2010b). Interestingly, either electrical (Kim et al., 2010b) or pharmacological (Chang and Maren, in press) activation of the prefrontal cortex were shown to alleviate the immediate extinction deficit. Although recent fear is resistant to extinction, interventions targeting enhancement of medial prefrontal cortical activity may facilitate extinction, particularly under conditions in which it normally fails (Thompson et al., 2010). In sum,

although post-conditioning protein synthesis inhibition effectively impairs the consolidation of fear memory, immediate extinction does not.

Post-retrieval deconsolidation

The brief time window after acquisition that memory is susceptible to disruption produces logistical challenges for intervention. However, another temporal window in which fear memory is sensitive to disruption is shortly after retrieval (Misanin et al., 1968; Nader and Hardt, 2009a; Sara, 2000). Like new memories, older memories appear to become labile yet again once they are retrieved and reactivated (Figure 3). This suggests that consolidated fear memories might be vulnerable to disruption soon after they have been retrieved. Consistent with this possibility, Nader and colleagues have shown in an influential series of experiments that manipulations that interfere with the consolidation of fear memory also disrupt fear memory when administered shortly after retrieval of that memory (Nader et al., 2000). In these experiments, rats underwent standard auditory fear conditioning in which a tone CS is paired with a footshock US. The next day a single CS was presented to retrieve the fear memory, and this was followed immediately by an infusion of the protein synthesis inhibitor anisomycin into the BLA. Although anisomycin spared short-term retention of the fear memory, it severely impaired the long-term retention of that memory. Although debate continues about the nature of this deficit (Lattal and Abel, 2004; Miller and Matzel, 2000; Rudy et al., 2006), there is considerable evidence that it results from a failure to maintain the fear memory in a consolidated state.

This work suggests that memory consolidation is a dynamic process that is not unique to the encoding of new memory. In fact, memory retrieval appears to “deconsolidate” established memory traces returning them to a labile and destabilized state that requires protein synthesis-dependent reconsolidation for long-term retention. The mechanisms of deconsolidation are not known, and it is unclear whether memory reactivation actually reverses the outcome of consolidation or renders the consolidated trace labile in some other way. In either case, interfering with reconsolidation after retrieval leads to memory loss: the deconsolidated memory fails to stabilize and decays much as short-term memory decays in the absence of consolidation to long-term memory (Figure 3). Although reconsolidation has been described in many memory systems, it is bounded (Nader, 2006). For example, the sensitivity of reactivated memories to protein synthesis inhibitors is related to many factors including the age and strength of the memory (Milekic and Alberini, 2002; Wang et al., 2009). In addition, not all forms of memory appear to undergo protein synthesis-dependent reconsolidation (Nader and Hardt, 2009b). Nonetheless, the sensitivity of long-term fear memories to retrieval-based manipulations provides a much more tractable time window for therapeutic intervention insofar patients with anxiety disorders often seek treatment long after trauma.

As a consequence, several groups have attempted to disrupt consolidated fear memories by interfering with reconsolidation processes after reactivation. Because there is strong interest in developing effective interventions for patients with anxiety disorders, the focus has been on developing interventions that can be safely administered to humans. For example, in rats systemic administration of the beta-adrenergic receptor antagonist, propranolol, disrupts the reconsolidation of fear memories under some conditions (Debiec and LeDoux, 2004; Muravieva and Alberini, 2010). A pair of studies in humans similarly suggests that propranolol administration can influence the reconsolidation of fear memory. In one report (Kindt et al., 2009), healthy subjects underwent a fear-potentiated startle conditioning procedure followed by oral propranolol administration and memory reactivation the day after conditioning. Interestingly, propranolol disrupted the retention of one index of fear memory (i.e., the conditioned acoustic startle response), but spared declarative memory of the CS-US relationship (i.e., shock expectancy). This effect was not due to propranolol

administration alone, insofar as administering propranolol without reactivating the memory did not dampen startle. In other words, reactivating the fear memory after propranolol administration appeared to remove the emotional tone of the aversive experience while sparing the memories of the stimulus contingencies and the conditioning experience itself. In another study, patients with PTSD were given oral propranolol after recalling events related to their trauma (Brunet et al., 2008; Pitman et al., 2006). One week later physiological responses to those trauma-relevant memories were assessed. Relative to placebo controls, patients administered propranolol exhibited lower heart rate and skin conductances when recalling trauma-related memories. It is not clear in this case, however, whether propranolol administration alone would produce a similar outcome (i.e., a non-reactivated propranolol group was not run). Nonetheless, these results suggest that pharmacological disruption of fear memory reconsolidation may be an effective intervention for reducing some indices of fear and anxiety.

In addition to pharmacological approaches to reducing fear memory, it has recently been argued that delivering extinction trials shortly after reactivation of fear memory might erase those memories. In these experiments, extinction trials were delivered from 10 minutes to an hour after reactivation of a fear memory conditioned twenty-four hours earlier (Monfils et al., 2009; Schiller et al., 2010). Under these conditions, the extinction of fear in the reactivated subjects did not exhibit renewal (Monfils et al., 2009), reinstatement (Monfils et al., 2009; Schiller et al., 2010), or spontaneous recovery (Monfils et al., 2009; Schiller et al., 2010); extinction in non-reactivated subjects exhibited recovery. Only one of the studies examined the duration of the effect, and in that case it was reported to last at least one year (Schiller et al., 2010). Hence, the failure of fear to recover under these conditions suggests that administering extinction trials during the reconsolidation window leads to a permanent disruption of the fear memory. This suggests that extinction can disrupt the reconsolidation of fear under some circumstances (e.g., soon after retrieval), and lead to loss of the fear memory itself. It should be noted, however, that the generality of this effect is not yet clear. McNally and colleagues recently examined post-reactivation extinction using procedures nearly identical to those used in the previous experiments (Chan et al., 2010). Unlike the previous reports, McNally and colleagues failed to observe impaired renewal and reinstatement in rats receiving extinction trials shortly after reactivation of the fear memory. In fact, there was a trend for more robust renewal when extinction was conducted after reactivation, suggesting that extinction after memory retrieval does not impair fear memories as previously proposed. Clearly, further work is necessary to understand the conditions under which extinction training yields impairments in long-term fear memory.

However, given the considerable therapeutic potential of eliminating fear memories, there is obvious interest in defining the molecular processes governing reconsolidation (Tronson and Taylor, 2007). In fact, recent work indicates that metabotropic glutamate receptors (mGluR1) receptors may confer susceptibility of fear memories to disruption by extinction (Clem and Huganir, 2010). In these studies, mice that underwent fear conditioning were found to exhibit significant increases in AMPA-receptor mediated synaptic transmission in the lateral amygdala *in vitro*. A portion of this increased AMPA-receptor mediated current was maintained by AMPA receptors lacking GluA2 receptors, which are also calcium-permeable (CP-AMPA receptors). Interestingly, low-frequency electrical stimulation of LA synapses induced mGluR1-dependent long-term depression (LTD) that was mediated by a reduction in CP-AMPA receptor-mediated current. Similarly, delivering extinction trials 30 minutes after reactivation of the fear memory also led to a decrease in CP-AMPA receptor-mediated current in the LA *in vitro* and reduced the recovery of fear that normally occurs after extinction. The behavioral effect of post-retrieval extinction was impaired after systemic administration of an mGluR1 antagonist, suggesting that mGluR1-mediated synaptic depression mediates fear memory erasure. NMDA receptors also play a role in

inducing synaptic depression, and previous work indicates that NMDA antagonists also prevent fear memories from becoming labile after a reminder (Ben Mamou et al., 2006).

A critical remaining question, however, is why a CS reminder was required to deconsolidate LA synapses to render them susceptible to mGluR1-mediated removal of CP-AMPA receptors in the first place. Others have reported that extinction *without memory reactivation* also yields mGluR1- and NMDA-dependent depotentiation of lateral amygdala synaptic transmission and a reduction in the surface expression of GluA1- and GluA2-containing AMPA receptors (Kim et al., 2007). In fact, consolidated, rather than labile, memories appear to be more sensitive to GluR1-mediated depotentiation (Kim et al., 2010a). Clearly, the regulation of AMPA receptor expression in the lateral amygdala is involved in both the acquisition and extinction of fear, but the behavioral modulation of AMPA receptor endocytosis after fear conditioning is poorly understood. Nader and colleagues have shown that NMDA receptors are required for a CS reminder to render fear memory sensitive to subsequent protein synthesis inhibition (Ben Mamou et al., 2006), so it is conceivable that an NMDA-receptor dependent process induces susceptibility to mGluR1-mediated LTD involved in reversing conditioning-related changes in the LA. Yet how particular behavioral experiences confer susceptibility of synapses to AMPA receptor endocytosis is unknown.

It is noteworthy that the sensitivity of fear memories to “reconsolidation update” (i.e., extinction after reactivation) appears to be time-limited. For example, Clem and Haganir (2010) found that fear memory erasure and CP-AMPA receptor removal was maximal 24 hours after conditioning, reduced 48 hours after conditioning, and not possible one week after fear conditioning. It has also been suggested that the passage of time limits the sensitivity of fear memories to protein synthesis inhibition after reactivation (Anokhin et al., 2002; Milekic and Alberini, 2002). Collectively, these results reveal that memories at the earliest stages of consolidation are the most sensitive to disruption, whether by post-conditioning or post-retrieval protein synthesis inhibitors or post-retrieval extinction manipulations. A continued challenge is how to lengthen the window of susceptibility such that even the most enduring fear memories can be eliminated.

Erasure

Preventing the reconsolidation of fear memory leads to a reduction in fear behavior, but there is some debate about the nature of this impairment. On the one hand, many authors have found that post-retrieval manipulations yield a non-recoverable loss of performance, suggesting that destabilized memory traces vanish if they are not reconsolidated. On the other hand, others have found that performance impairments after these manipulations are transient, suggesting that temporary retrieval failures, rather than disruption of the memory trace *per se*, underlie the effects of post-retrieval manipulations of memory (Lattal and Abel, 2004; Power et al., 2006). Indeed, it is perhaps not surprising that reactivation approaches would spare at least some aspects of the original memory insofar as the typical reactivation procedure may not retrieve the entire memory (Debiec et al., 2006; Doyère et al., 2007). Failing to reactivate the entire associative network of a memory might protect that memory from the influence of post-retrieval manipulations. In essence, complete erasure of a memory would require that the entire associative network containing that memory be eliminated.

To this end, Josselyn and colleagues have made use of an innovative molecular genetic approach to recruit and then disable a network of neurons in the amygdala mediating conditioned fear (Han et al., 2009). To recruit a network of amygdala neurons during fear conditioning, they used a viral vector to overexpress CREB, a transcription factor previously shown to bias amygdala neurons for inclusion in the neural network underlying fear memory (Han et al., 2007). To selectively target these neurons, they used transgenic mice (iDTR)

that express the simian diphtheria toxin receptor under the control of Cre-recombinase (cre). In these mice, infusion of a replication-deficient herpes simplex virus expressing CREB-cre into the lateral amygdala renders neurons overexpressing CREB sensitive to apoptosis by systemic injection of diphtheria toxin. In an elegant series of experiments, Josselyn and colleagues found that ablating CREB-cre neurons recruited during fear conditioning severely and selectively impaired the expression of fear memory. The deficits in fear memory were long lasting, and several control experiments ruled out the possibility that non-specific damage to the amygdala contributed to the memory impairments. In fact, the most parsimonious interpretation of these results is that the investigators selectively erased the neuronal network in the amygdala harboring the memory trace.

Another approach to erasing memory targets the molecules within neurons that maintain long-term memories. Although there are several candidate molecules involved in memory maintenance (Kandel, 2009; Martin et al., 2000), one molecule in particular has received considerable attention as a substrate for long-term memory (Sacktor, 2011). Protein kinase M zeta (PKMzeta), which is a constitutively active isoform of protein kinase C, is involved in both the maintenance of synaptic long-term potentiation (Ling et al., 2002; Osten et al., 1996) as well as several forms of learning and memory (Pastalkova et al., 2006; Sacktor, 2011; Serrano et al., 2008). Within the amygdala, for example, it has been shown that inhibition of PKMzeta with a pseudosubstrate of the kinase (zeta inhibitory peptide or ZIP) impairs the expression of consolidated fear memories (Kwapis et al., 2009; Migues et al., 2010; Serrano et al., 2008). Recent data suggest that ZIP impairs memory by interacting with GluA2-containing AMPA receptors in the amygdala. Like CP-AMPA receptors (that lack GluA2), GluA1/2 receptors appear to be driven into LA synapses after fear conditioning (Kim et al., 2007; Mao et al., 2006; Rumpel et al., 2005) and PKMzeta appears to have a role in maintaining the surface expression of these receptors after learning (Migues et al., 2010). The precise regulation of GluA2-lacking and GluA-2 containing AMPA receptors is likely to be quite complex. Nonetheless, it appears that both types of glutamate receptors are upregulated at amygdala synapses after fear conditioning and pulling down either class of receptor after learning influences the retention of fear memories.

Turning back the clock

Clearly, the stability of fear memory represents presents a major challenge to manipulations designed to eliminate fear memories. But are fear memories necessarily resistant to erasure? Recent studies on the ontogeny of fear extinction have provided some interesting insight into the stability of fear memory across the lifespan. Recent studies by Richardson and colleagues have examined whether age influences the properties of extinction in rats (Kim and Richardson, 2007, 2008, 2010). Like adults, recently weaned 23-day old exhibit both contextual and auditory fear conditioning and extinction of that fear exhibits renewal, reinstatement, and spontaneous recovery. Surprisingly, however, 17-day old preweanling rats exhibited an unusual form of extinction that does not exhibit any of the hallmark recovery phenomena (e.g., renewal, reinstatement, and spontaneous recovery) that are associated with extinction in older rats. In other words, extinction may erase conditioned fear in preweanling rats. Interestingly, preweanling rats do not exhibit contextual fear conditioning (Rudy and Morledge, 1994), suggesting that they may also have global deficits in contextual processing functions required for renewal, spontaneous recovery, and reinstatement. Indeed, the nature of the renewal deficits in young rats is similar to that in adult rats with hippocampal lesions (Corcoran et al., 2005; Corcoran and Maren, 2001, 2004; Hobin et al., 2006; Ji and Maren, 2008a; Ji and Maren, 2005): both young rats and adult rats with hippocampal lesions fail to renew fear to an extinguished CS outside of the extinction context. Hence, the development of the hippocampus may afford a flexible memory system that allows a CS to mean different things in different contexts.

This explanation of how extinction comes to be resistant to erasure emphasizes the development of neural systems that allow the flexible representation of information. Another possibility is that the synaptic network that encodes fear memory is fundamentally different in young animals. In fact, Gogolla and colleagues (2010) have recently observed in mice that the development of the amygdala extracellular matrix, in particular perineuronal nets composed of chondroitin sulfated proteoglycans (CSPGs), parallels the development of extinction learning (Gogolla et al., 2009). Interestingly, infusion of a CSPG-degrading enzyme (chondroitinase ABC or chABC) into the amygdala of an adult mouse digested perineuronal nets and produced a fear extinction phenotype like that of a young mouse. That is, chABC-treated mice exhibited normal conditioning and consolidation of fear conditioning, and also showed normal decreases in conditioned fear after extinction training. Remarkably, however, chABC-treated rats did not show spontaneous recovery or renewal of fear. This suggests that perineuronal nets, while not necessary for the acquisition of fear memory, may prevent those memories for destabilizing after extinction training. The mechanism by which degradation of perineuronal nets alters the stability of fear memory is not known, although chABC impairs several forms of synaptic plasticity in the amygdala and hippocampus. Independent of the precise mechanism, however, these data suggest that molecular factors at the synapse are not only involved in the long-term maintenance of memory, but in protecting those memories from the destabilizing influences of other behavioral experiences. Unfortunately, though, removing perineuronal nets in the amygdala only promotes a non-recoverable extinction of fear when chABC is applied before, but not after fear conditioning. This obviously limits the therapeutic potential of compounds targeting perineuronal nets insofar as they would have to be administered before a traumatic experience, and would presumably be ineffective at promoting the suppression of old fears.

Conclusions

After decades of research aimed at how fears are learned, the last five years have witnessed an explosion of work on the neural mechanisms underlying how fear memories are suppressed and, ultimately, erased. Although fear memories are typically long-lived, there is now considerable evidence that they can be erased under some conditions. For instance, pharmacological disruption of molecules critical for memory reconsolidation and memory maintenance produce enduring fear loss. Moreover, behavioral manipulations, such as extinction, appear to yield fear erasure under some circumstances. These phenomena provide insight not only into the mechanisms underlying the encoding and regulation of fear and extinction memories, but also illuminate novel clinical interventions in patients with pathological fear memories.

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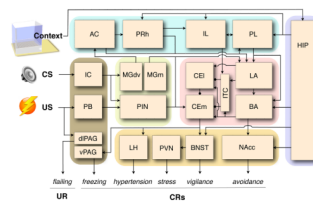


Figure 1.

Neuronatomy of conditioned fear. A simplified neuroanatomical schematic outlining some of the major brain regions and their anatomical connections involved in Pavlovian fear conditioning. Shading indicates major brain regions (brown, midbrain; yellow, thalamus; orange, basal forebrain; blue, neocortex; red, amygdala; violet, hippocampus). Sensory information ascends to the amygdala through the midbrain and thalamus; auditory information also reaches the amygdala via the cortex. Anatomical convergence and association of the conditioned stimulus (CS) and unconditioned stimulus (US) occurs in the amygdala; contextual information processed by the hippocampus can also enter into association with the US in the amygdala. Conditioned and unconditioned fear responses (CRs and URs) are mediated by projections from the amygdala to an array of brain areas involved in autonomic and somatic defensive responses. Abbreviations: AC, auditory cortex; BA, basal nuclei of the amygdala; BNST, bed nuclei of the stria terminalis; CEl, lateral division of central nucleus of the amygdala; CEm, medial division of the central nucleus of the amygdala; HIP, hippocampus; IC, inferior colliculus; IL, infralimbic division of the medial prefrontal cortex; ITC, intercalated cells of the amygdala; LA, lateral nucleus of the amygdala; LH, lateral hypothalamus; MGdv, dorsal and ventral divisions of the thalamic medial geniculate nucleus; MGm, medial division of the thalamic medial geniculate nucleus; NAcc, nucleus accumbens, dIPAG, dorsolateral division of the periaqueductal gray; vPAG, ventral division of the periaqueductal gray; PIN, posterior intralaminar nucleus of the thalamus; PL, prelimbic division of the medial prefrontal cortex; PRh, perirhinal cortex; PVN, paraventricular nucleus of the hypothalamus.

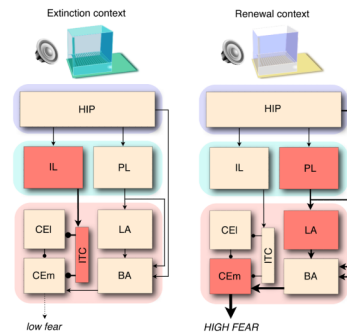


Figure 2.

Contextual control of extinction. A simplified neuroanatomical schematic illustrating differential c-fos expression in brain structures involved in the expression of fear after extinction. Suppression of fear to a CS in the extinction context (left) is associated with activity in the infralimbic division of the medial prefrontal cortex (IL) and inhibitory intercalated neurons (ITC) in the amygdala. Inhibition of the medial division of the central nucleus (CEm) by the ITC limits the expression of conditioned fear. The return of fear to a CS presented outside of the extinction context (Renewal context, right) is associated with activity in the prelimbic division of the medial prefrontal cortex (PL), the lateral nucleus of the amygdala (LA), and CEm. The hippocampus (HIP) basal nuclei of the amygdala (BA) are engaged in both situations, and may therefore gate the expression of conditioned fear through their connections with the the unique prefrontal-amygdala networks associated with extinction and renewal.

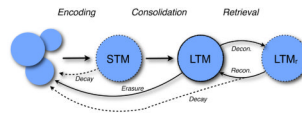


Figure 3.

Stabilization and destabilization of memory. Memory formation is associated with initial encoding to establish a short-term memory (STM) followed by a time-dependent consolidation phase to establish a stable long-term memory (LTM). Retrieval of LTM destabilizes or deconsolidates (decon.) the memory (LTM_r) rendering it labile once more; the persistence of LTM after retrieval requires reconsolidation (recon.). Failure to reconsolidate the memory trace results in decay, much as STM decays in the absence of consolidation. In the absence of retrieval, LTM may be actively erased by a variety of manipulations by interfering with the molecular mechanisms involved in memory maintenance.